Write shell scripts: Bash is not only an interactive command prompt but is also a powerful scripting language interpreter. As many terminal tasks require more than one shell command, you can store them in a shell script text file and pass the variable data as command-line arguments. I use the following script to kill applications by their name:

```
'for sID in pgrep $1'
do
      ps $sID
      kill -STOP $sID
done

for sID in 'pgrep $1'
do
      ps $sID
      kill -KILL $sID
done
```

To kill all instances of the Opera browser from the command line, I can type *bash ~/MyScripts/kill.txt opera*. Even though an *execution bit* can be added to the file so that bash can be omitted from the command, I never do this. As a long-time Windows user, I am paranoid about malware finding executable scripts and adding their own destructive commands. In Microsoft Windows operating systems, I usually change the default action of *.vbs*, *.js*, *.wsf* and *.reg* file types to *Edit*, rather than leave them at *Open* or *Merge*. I follow the same precaution in GNU/Linux.

Creating command aliases: You can create abbreviated forms for your commands in the *.bashrc* file. One of the most common commands that you have to type may be *ls l*. Hence, I have the following alias for the *ls* command. I just need to type one letter (l) for it:

```
alias l= 'ls l'
```

To execute my 'kill' script, I use the following alias:

```
alias kll= 'bash ~/MyScripts/killit.txt'
```

So, to kill Opera, I type 'kll opera'. I continue to use the obsolete pre-Blink Opera browser for its wonderful RSS reader. Unfortunately, it becomes unstable several times during a session. It cannot be killed by the usual 'killall' or 'pkill' command. The kill script and this alias, however, vanquishes it.

Using bash, not sh: My first experience with a UNIX-like OS was on SCO UNIX. As a result, I was accustomed to running shell script files with the *sh* command. This continued even after I started using GNU/Linux systems. For years, I was perplexed by why many of the scripts were not working well, but they did fine after I set the execution bit ('chmod +x') and ran them directly without the shell command. Apparently, in many GNU/Linux distributions, *sh* refers to the

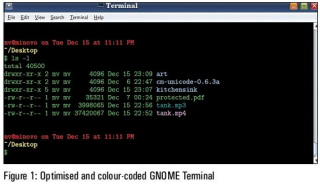


Figure 1: Optimised and colour-coded GNOME Terminal

old UNIX-like shell and bash is a separate program that works like a more advanced superset of 'sh'. As a result, scripts designed for bash will not work well with *sh*. So, always use bash to run all your bash shell scripts. However, there are some tasks such as cross-platform driver compilation that still require *sh*. Another thing to note is that the *su* or *roo*t terminal uses *sh* by default. This is why the up/down arrow keys will not let you browse the history. So, be aware of which shell program you are currently using.

Breaking up long commands: When you have to type a long command, you can type a 'slash' to easily wrap the command to the next line and continue from there, as follows:

```
ffmpeg -i tank.mp4 \
       -c:v copy   \
       -c:a copy   \
       -ss 1:12    \
       -t 2:50     \
       tank-cut.mp4
```

Using keyboard shortcuts: GNOME Terminal is a multi-tab window. You can press *Ctrl+Shift+T* and start a terminal in a new tab in the same window. If you feel you need to refer to the man pages of the current command, do not cancel what you are typing. Just start a new tab and browse the man pages from there. Another useful shortcut is *Ctrl+U*, which cancels or deletes everything in the current line and lets you begin afresh. To copy text from a terminal window, the short cut is *Ctrl+Shift+C*, and the short cut to paste text is *Ctrl+Shift+V*.

Ctrl+C is for killing the current command and should not be used to copy text in the terminal.

Switching to a TTY terminal: The Linux desktop managers are known to be very stable. However, on those rare occasions when the desktop hangs, you can hold down *Ctrl+Alt* and press any of the function keys to switch to a 'TTY terminal'. These terminals are created by the OS before loading the GUI (desktop manager). From one of these terminals, you can log in to your account and do your troubleshooting. One of the following commands can restart the desktop manager.

```
sudo service gdm restart # for Gnome 2
sudo service mdm restart # for Mate
```